MOSAIC

Modular Open Source Architecture In Cabin

An open standard for interoperable aircraft cabin systems

TECHNICAL OVERVIEW · 2026

CONTENTS

Agenda

01

Introduction

Modular avionics today and the case for open integration

Modular hardware is mature — integration is not

Long history. Modular avionics have been a part of aircraft systems for more than 50 years — for example, ARINC 404 and 600 rack-mount trays.

Hardware modular, systems aren't. Most modern aircraft adopt modular hardware design but still integrate systems only through IP or proprietary protocols.

Safety drives silos. The inherent need for aircraft safety and security pushes companies to silo communication protocols, technical issues and intellectual property.

50+

years of modular avionics hardware

...yet cabin subsystems still rarely speak a common language.

Building on the MOSA precedent

2017

MOSA memo

A joint military memorandum directs a Modular Open Systems Approach (MOSA).

2021

Codified in law

Legislation requires a Modular Open Systems Approach for military weapon systems.

Now

MOSAIC

Extends the MOSA framework to aerospace cabin interfaces — initially for Part 91 and 135 operations.

Consumer momentum. The rapid arrival of IoT (Internet of Things) devices in the consumer market has driven a more agile, collaborative ecosystem — exactly the model MOSAIC brings to the cabin.

02

Current System Architecture

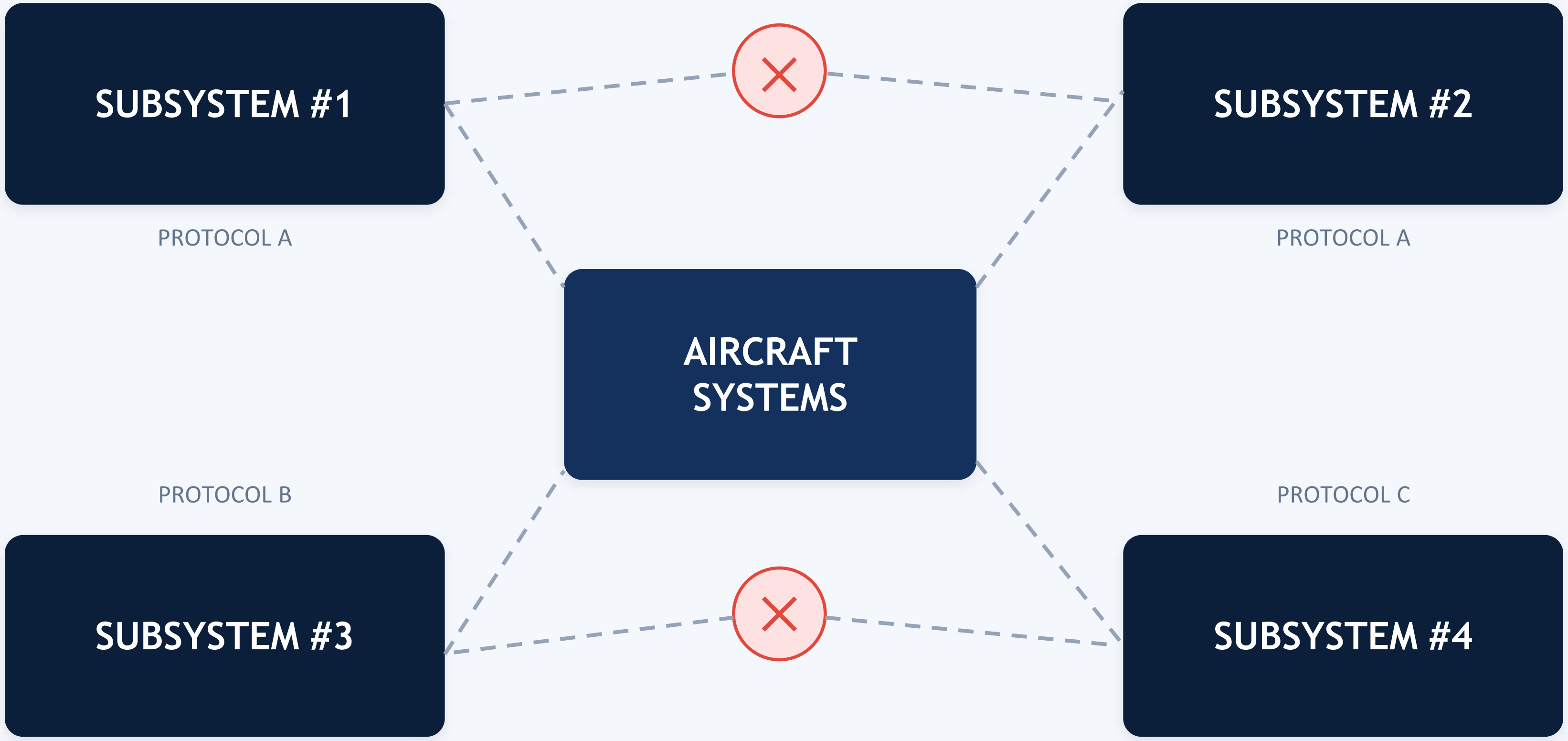
How cabin subsystems are integrated today

Many subsystems, no common language

1. Cabin systems in modern business jets contain several subsystems.
2. Most of these systems do not interact naturally.
3. Even when built by the same vendor, subsystems do not always follow a standard set of communication protocols or integration methods.
4. Sustainability is lost when vendors are absorbed, go defunct, or create siloed proprietary communication methods.
5. Integrators and OEMs repeat NRE and over-complex architectures to solve simple problems — like turning the lights on and off.

Subsystems can't natively interoperate

Subsystems are built by different vendors and interfaced in the aircraft cabin — but they speak different protocols.



No common protocol → no native interoperability

Third-party bridges fill the gaps

Where communication is not common, third-party solutions or additional equipment become necessary.

Every mismatch adds hardware, cost and another vendor to manage.

The path runs through an intermediary

The communication path requires an intermediate solution to translate between subsystems.

SUBSYSTEM #1 → TRANSLATOR / GATEWAY → INTEGRATOR LOGIC → SUBSYSTEM #4

Indirection means more points of failure, more integration effort, and proprietary links that aren't freely available.

Rigid, proprietary, and integrator-dependent

Integrator-dependent	Current systems rely heavily on the integrator or OEM to bridge communication paths — often introducing additional vendors to close the gaps.
Closed by default	A vendor's proprietary protocol is usually not freely available, locking integration behind licensing restrictions.
Rigid by design	The structure is intentionally rigid, providing safety and security for the overall subsystems — but at the cost of flexibility.

03

MOSAIC Overview

Defining the open, modular cabin standard

What MOSAIC is

WHAT

MOSAIC

- MOSAIC — Modular Open Source Architecture In Cabin.
- Defines an open, modular set of standards and interfaces between aircraft cabin systems and sub-systems.
- Promotes non-proprietary communication between systems to improve innovation, data sharing and reduce complexity.
- Hosts a community of stakeholders to encourage collaboration, fuel best practices and provide centralized training.

Why MOSAIC exists

WHY

MOSAIC

- Enable systems and software applications to access and provide data and services using open interface definitions between components — adopted from MOSA.
- Provide a sustainable approach to aircraft cabin design that allows for innovation, acceleration and improved integration.

Open interfaces turn one-off integration work into reusable, shareable building blocks.

Who drives MOSAIC

WHO

MOSAIC

- MOSAIC principles should be driven by aircraft cabin integrators and OEMs.
- Continued open collaboration with vendors and system stakeholders is essential.
- MOSAIC is being developed under an open-source license (to be finalized).

Where MOSAIC applies

WHERE

MOSAIC

- MOSAIC is not intended to change system design.
- Like MOSA, it focuses on inter-communication and integration — system to system, and system to sub-system.
- MOSAIC targets cabin systems only in the near term.

Scope is deliberately narrow: the interfaces between systems, not the systems themselves.

04

Framework

The blueprint, the approach, and the benefits

A blueprint for cabin architectures

A shared blueprint

MOSAIC acts as a blueprint for aircraft cabin system architectures and products.

100% inclusive

Built to include every stakeholder — OEMs, integrators, vendors and the supply chain.

Defined by modularity

The framework is defined through modularity and well-bounded interfaces.

Not all-or-nothing

Partial adoption still wins: 50% cross-modularity is better than 0%.

The approach — adopted from MOSA

01

Modular Design

Decompose the cabin into discrete, well-bounded modules.

02

Defined Interfaces

Specify how modules interact — formats, protocols, error handling.

03

Accessible Data

Make data reachable and shareable across components.

04

Open Interfaces

Require open boundaries so third parties can integrate.

Five benefits — adopted from MOSA

- Innovation
- Competition
- Interoperability
- Cost Savings / Avoidance
- Tech Refresh

BENEFIT 01 / 05

Innovation

- Take advantage of new advancements in technology.
- Enable technical agility to meet rapidly changing requirements.

BENEFIT 02 / 05

Competition

- Platform and vendor independence when hardware and software implement open industry standards.
- Ability to openly compete severable modules.
- Compete portable components against open specifications and standards for interfaces, services and supporting formats across a wide range of systems.

BENEFIT 03 / 05

Interoperability

- Operational flexibility to support reconfigurable product configurations of existing capabilities — to counter threats or enable different missions.
- Share and exchange data consistently between components and stakeholders using defined data models.

BENEFIT 04 / 05

Cost Savings & Avoidance

- Achieve less expensive technical modifications.
- Add capabilities and modifications without redesigning non-critical hardware or software.
- Reuse previous investments — technology, modules and components — across the acquisition lifecycle.

BENEFIT 05 / 05

Tech Refresh

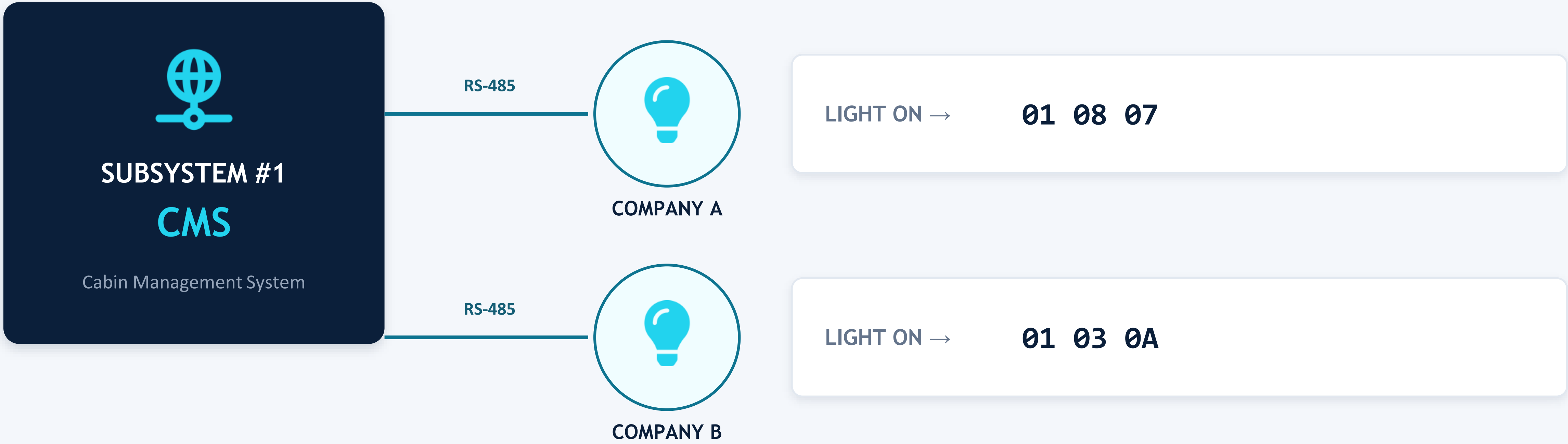
- Enable technical flexibility for rapid and effective system upgrades.
- Upgrade technology without changing every component in the entire system.

05

Examples

From proprietary commands to open APIs

Two vendors, two command sets



Two vendors supply lighting. Each defines its own command architecture, so the CMS must integrate two separate command logics for the same RS-485 link.

What are the problems?

NRE	The CMS provider invests non-recurring engineering to integrate both lighting vendors — a cost typically burdened on the OEM or completion center.
Sustainability	If the CMS provider goes out of business, a replacement must re-integrate the existing lights — and may charge fresh NRE to pair a new CMS with older lighting.
Timing	Integration schedules can leave a newly installed system already behind the consumer technology curve.

Complexity compounds the cost

Complexity

Multiple system installations create additional cost for the integrator or OEM — and every extra LRU adds weight, wiring and failure points.

- Reduce LRU count — save weight and cost.
- Build workforce confidence and familiarity.
- Let engineers focus on product features.
- Give installers and maintainers common training and support paths.

Flexibility and shared knowledge

Flexibility

Future integrations become cost-effective without ripping out whole subsystems — and upgrades can be software-defined.

Commonality

Sharing problems and fixes across the community on open communication paths drives faster resolution across the entire integrator network.

Why this matters

Simplicity

Fewer control architectures means lower integration complexity, cost and training.

Sustainability

Systems become scalable — ready for future replacement or upgrade to a known control method.

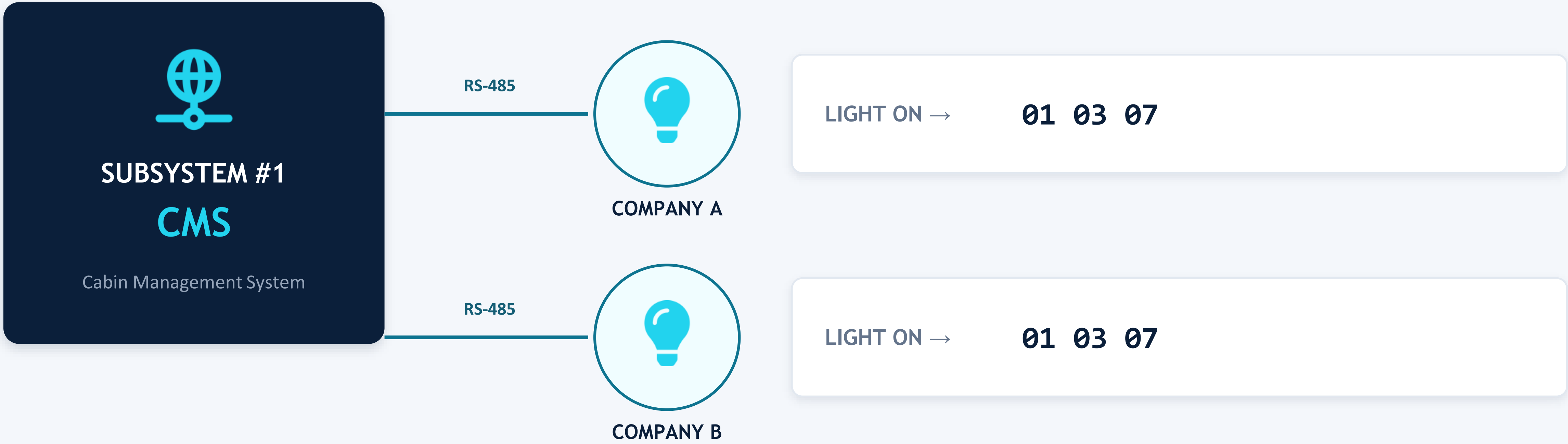
Competition

Vendors compete on an even field — design, MTBF, weight — letting small players challenge large ones and large players stay flexible.

Cost

Common standards reduce cost for every stakeholder.

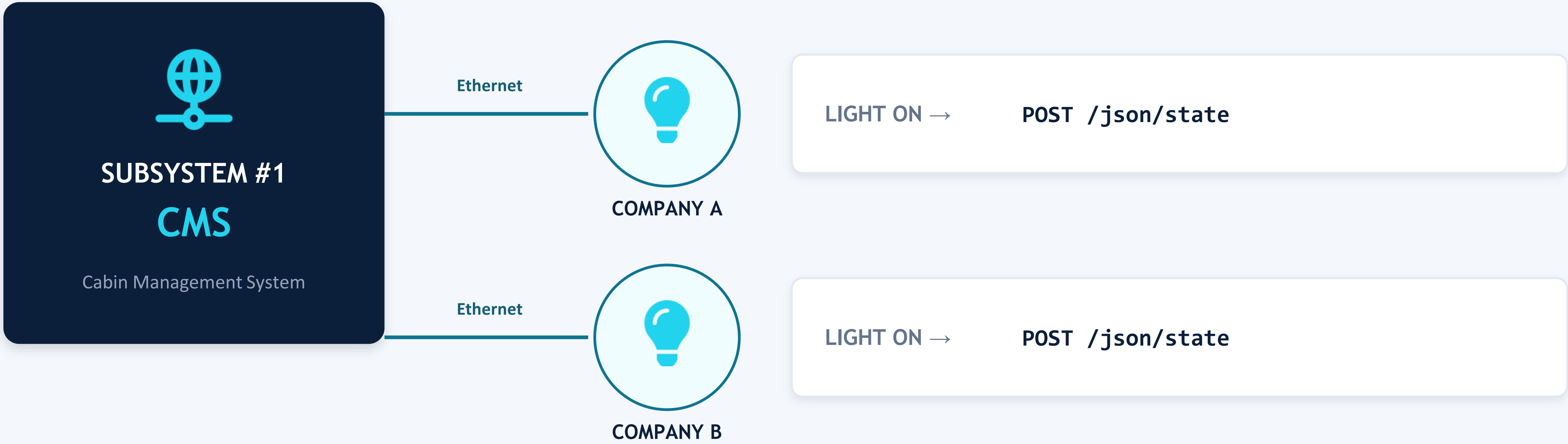
Step 1 — standardize the command



With a standard command architecture in place, the circuit becomes easier to integrate, troubleshoot and scale. In some cases the communication method itself can be simplified.

EXAMPLES

Step 2 — move to an open API



The communication method is changed to an IP (802.x) method, enabling full system and sub-system integration. Both vendors now use the same open MOSAIC API to control the light.

06

The Path

Working group, specification and roadmap

Where MOSAIC is headed

- **Kickoff** — Initial exploratory group of vendors, completion centers and a VIP fleet operator at AIX 2024.
- **First working group** — First official working-group meeting held at NBAA 2024.
- **Specification** — In work — hosted publicly on GitHub for now.
- **Working group target** — X3 Completion Centers, 8 core vendors across all segments (CMS, Lighting, Audio)
- **Aircraft** — Completed: 1 (Collins core), 1 (Alto core), 3 (Nargoyle core), plus several modification aircraft as of June 2026

References

Modular Open Systems Approach (MOSA) Reference Frameworks in Defense Acquisition Programs
https://www.cto.mil/wp-content/uploads/2023/06/MOSA-Framework-2020.pdf

Defense Standardization Program — MOSA
https://www.dsp.dla.mil/Programs/MOSA/

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MOSAIC Working Group
www.mosaic-aero.com